# Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

**AY: 2025-26**

| Class: | TE | Semester: | VI |
|---|---|---|---|
| Course Code: | CSL601 | Course Name: | Data Analytics and Visualization Lab |

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|---|---|
| **Name of Student:** | **Shravani Sandeep Raut** |
| **Roll No.:** | **40** |
| **Experiment No.:** | **06** |
| **Title of the Experiment:** | **Implementation of Sentiment Analysis in Python.** |
| **Date of Performance:** | **10/03/2026** |
| **Date of Submission:** | **17/03/2026** |

**Evaluation**

| Performance Indicator | Max. Marks | Marks Obtained |
|---|---|---|
| Performance | 5 | |
| Understanding | 5 | |
| Journal work and timely submission | 10 | |
| Total | 20 | |

| Performance Indicator | Exceed Expectations (EE) | Meet Expectations (ME) | Meet Expect Below Expectations (BE) |
|---|---|---|---|
| Performance | 4-5 | 2-3 | 1 |
| Understanding | 4-5 | 2-3 | 1 |
| Journal work and timely submission | 8-10 | 5-8 | 1-4 |

**Checked by**

**Name of Faculty: Ms. Odilia Gonsalves**

**Signature:**

**Date:**

# Experiment No. 6

**Aim:** Implementation of Sentiment Analysis in python.

**Objective: -** To understand the use of various sentiment Analysis techniques by implementing them

**Description:**

**Sentiment Analysis:** Sentiment Analysis is a text analysis technique that allows companies to make sense of qualitative data. By detecting positive and negative sentiment in text data, such as tweets, product reviews, and support tickets, you can understand how customers feel about your brand, product, or service, and gain insights that lead to data-driven decisions

Sentiment Analysis deals with analyzing emotions and the perspective of a speaker or an author from a given piece of text. "Sentiment analysis or opinion mining refers to the appliance of language process, linguistics, and text analytics to spot and extract subjective information in supply materials". This field of technology deals with analyzing and predicting the hidden information keep within the text. This hidden information gives valuable insights regarding user's intentions, style and odds. Sentiment Analysis specializes in categorizing the text at the extent of subjective and objective nature. Judgement indicates that the text bears opinion content where's perspicacity indicates that the text is while not opinion content

**Some examples-**

1. **Subjective**- This motion picture by tom cruise and Angelina jolie is great. (This sentence has an opinion; it talks regarding the motion picture and also the writer's emotions regarding same "great" and thence its subjective

2. **Objective-** This motion picture stars tom cruise and Angelina.

(This sentence may be a reality, general information instead of an opinion or a read of some individual and thence its objective)

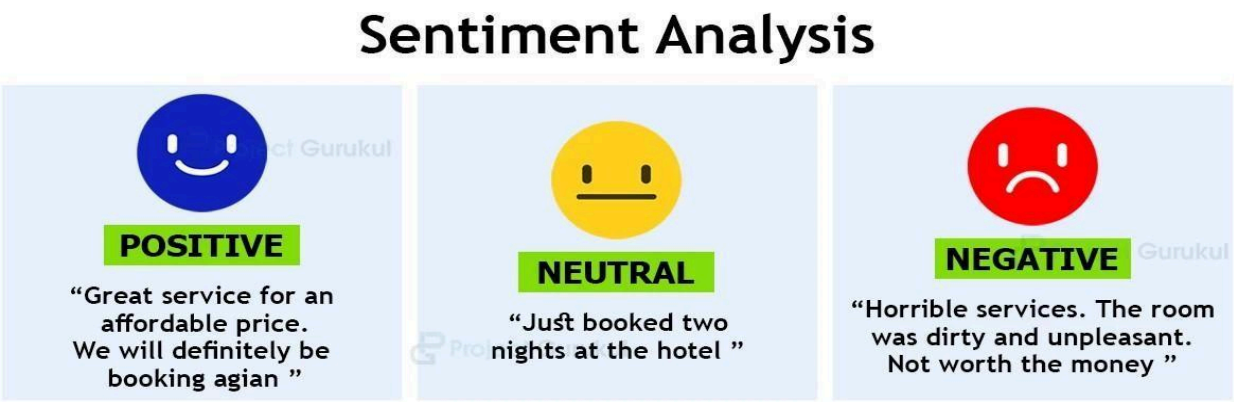
The subjective text may be additional categorized into three broad classes supported the emotions expressed within the text.

1. Positive- I like to look at Star series.

2. Negative- The movie was awful.

**3.** **Neutral-** I typically get hungry by evening. (This sentence has users views, emotions hence it's subjective however because it doesn't have any positive or negative polarity therefore it's neutral).

**Example:**




**Method 1: Using Positive and Negative Word Count – With Normalization for Calculating Sentiment Score**

Use Amazon Cell Phone Reviews dataset from Kaggle. **for Calculating Sentiment Score**

**Program:**

```
import pandas as pd
import re

file_path = "DAV/20191226-reviews.csv"
data = pd.read_csv(file_path)

reviews = data['body'].fillna("").astype(str)

positive_words = ["good","great","excellent","amazing","love","best","perfect"]
negative_words = ["bad","worst","poor","terrible","issue","problem","broken"]
```

```
print("--- METHOD 1: NORMALIZATION OUTPUT ---")
for i in range(5):
    text = reviews[i]
    words = re.findall(r'\w+', text.lower())
    total_words = len(words)

    pos = sum(1 for w in words if w in positive_words)
    neg = sum(1 for w in words if w in negative_words)
    score = (pos - neg) / (total_words) if total_words > 0 else 0

    print(f"Review {i+1}: {text[:50]}...")
    print(f"Total Words: {total_words} | Sentiment Score: {round(score, 4)}\n")
```

**Output:**

```
--- METHOD 1: NORMALIZATION OUTPUT ---
Review 1: I had the Samsung A600 for awhile which is absolut...
Total Words: 325 | Sentiment Score: 0.0062

Review 2: Due to a software issue between Nokia and Sprint t...
Total Words: 135 | Sentiment Score: -0.0148

Review 3: This is a great, reliable phone. I also purchased ...
Total Words: 133 | Sentiment Score: 0.0075

Review 4: I love the phone and all, because I really did nee...
Total Words: 109 | Sentiment Score: 0.0183

Review 5: The phone has been great for every purpose it offe...
Total Words: 130 | Sentiment Score: 0.0077
```

**Method 2: Using Positive and Negative Word Count – With Semi Normalization for Calculating Sentiment Score**

**Program:**

```
import pandas as pd
import re

file_path = "DAV/20191226-reviews.csv"
data = pd.read_csv(file_path)

reviews = data['body'].fillna("").astype(str)

positive_words = ["good","great","excellent","amazing","love","best","perfect"]
negative_words = ["bad","worst","poor","terrible","issue","problem","broken"]

print("--- METHOD 2: SEMI-NORMALIZATION OUTPUT ---")
for i in range(5):
    text = reviews[i]
    words = re.findall(r'\w+', text.lower())

    pos = sum(1 for w in words if w in positive_words)
    neg = sum(1 for w in words if w in negative_words)
    score = (pos - neg) / (pos + neg) if (pos + neg) > 0 else 0

    print(f"Review {i+1}: {text[:50]}...")
    print(f"Pos: {pos}, Neg: {neg} | Sentiment Score: {round(score, 4)}\n")
```

**Output:**

```
--- METHOD 2: SEMI-NORMALIZATION OUTPUT ---
Review 1: I had the Samsung A600 for awhile which is absolut...
Pos: 3, Neg: 1 | Sentiment Score: 0.5

Review 2: Due to a software issue between Nokia and Sprint t...
Pos: 0, Neg: 2 | Sentiment Score: -1.0

Review 3: This is a great, reliable phone. I also purchased ...
Pos: 1, Neg: 0 | Sentiment Score: 1.0

Review 4: I love the phone and all, because I really did nee...
Review 4: I love the phone and all, because I really did nee...
Pos: 2, Neg: 0 | Sentiment Score: 1.0

Review 5: The phone has been great for every purpose it offe...
Pos: 1, Neg: 0 | Sentiment Score: 1.0
```

**Conclusion:**

**Method 1:**

This method highlights the pure intensity of emotion by ignoring neutral words, making it ideal for classifying reviews as strictly positive or negative regardless of their length.

**Method 2:**

This method measures sentiment density by comparing emotional words against the total word count, effectively identifying how "opinion-heavy" or factual a review is.